Restaurant Data Analysis using Python

Objective

The objective of this project is to perform data analysis on a restaurant dataset to extract meaningful insights. The focus is on understanding rating patterns, customer preferences, geographical distribution, and the impact of services such as table booking and online delivery on restaurant performance.

Dataset Description

The dataset contains detailed information about restaurants such as name, address, city, country code, latitude, longitude, cuisines offered, aggregate rating, price range, and service availability like table booking and online delivery. This data helps in analyzing both categorical and numerical aspects.

Data Cleaning and Preprocessing

Initially, the dataset was checked for missing values. Missing values in the 'Cuisines' column were replaced with 'Unknown' to maintain consistency. Duplicate rows were identified and removed to ensure data quality. The structure and data types of all columns were verified to ensure correct analysis.

Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset better. The distribution of aggregate ratings was visualized using a histogram, showing that most restaurants fall within a moderate rating range. Value counts and percentage distributions were calculated for better understanding.

Statistical Analysis

Basic statistical measures such as mean, median, minimum, maximum, standard deviation, and count were calculated for numerical columns. These measures provide a summary of the dataset and help identify trends in restaurant ratings.

Location and Geospatial Analysis

The distribution of restaurants across cities and countries was analyzed using value counts. A correlation analysis was performed between latitude, longitude, and aggregate rating to understand any relationship between location and ratings. Additionally, an interactive map was created using Folium to visualize restaurant locations geographically.

Cuisine Analysis

The most common cuisines were identified by counting the frequency of each cuisine type. This helps in understanding customer preferences and popular food categories.

Table Booking Analysis

The percentage of restaurants offering table booking was calculated. The average ratings of restaurants with and without table booking were compared, providing insights into whether table booking influences customer satisfaction.

Online Delivery Analysis

The percentage of restaurants offering online delivery was analyzed. Further, online delivery availability was compared across different price ranges. A bar chart was used to visualize how delivery services vary with pricing.

Price Range Analysis

The most common price range among restaurants was identified. The average rating for each price range was calculated to determine which price category performs best. The price range with the highest average rating and its corresponding color indicator were identified.

Feature Engineering

New features were created from existing columns to enhance analysis. These include the length of the restaurant name and address. Additionally, categorical variables such as table booking and online delivery were encoded into numerical values (0 and 1) for better usability in further analysis or modeling.

Conclusion

This project demonstrates how data analysis techniques can be used to extract meaningful insights from a dataset. It highlights the importance of data cleaning, visualization, and feature engineering. The findings suggest that service availability and pricing influence restaurant ratings, and geographical visualization provides better understanding of distribution.

Tools and Technologies Used

Python programming language was used along with libraries such as Pandas for data manipulation, Matplotlib for visualization, and Folium for map-based analysis.